

Week 3A Exercises: Functions & Loops

1. Implement a function that takes in one parameter (a number from 1-7) and returns the day of the week (1 is Monday, 2 is Tuesday, and so on).

Output:

Enter day number: 3

Day 3 is Wednesday

2. Make a function called `cube()` that takes a parameter called `number`. The function should return the cube of that number (i.e. `number*number*number`). Define a second function called `divisible_by_three()` that takes an argument called `number`. If that number is divisible by 3, perform the previous function `cube(number)` and return the result. If the number isn't divisible by 3, return `False`. Check if it works!!!!.

Output:

Enter number: 5

False

Enter number: 6

216

Enter number: 7

False

Enter number: 9

729

3. Write a program that will print hours from "0:00" to "19:00" with interval of 1 hour as follow:

Output

6:00

7:00

8:00

9:00

10:00

11:00

12:00

13:00

14:00

15:00

16:00

17:00

18:00

19:00

4. Modify the previous program to also print “take medicine” every 4 hours starting 8:00 as follow:

Output

6:00

7:00

8:00 take medicine

9:00

10:00

11:00

12:00 take medicine

13:00

14:00

15:00

16:00 take medicine

17:00

18:00

19:00

5. Implement a factorial() function. It takes an integer as a parameter and calculates its factorial. No guidance for this one, it’s especially challenging. Needs a loop.

Output

Enter number: 3

Factorial of 3 is 6

6. Use the `math.sqrt` function. Write a program that prints the following table using the `sqrt` function in the `math` module.

Output

Number Square root

0 0.00

2 1.41

4 2.00

6 2.45

8 2.83

10 3.16

12 3.46

14 3.74

16 4.00

18 4.24

7. Write a Python program to print the following pattern, using a nested for loop for example for `n=5`.

Output

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8. Do 7 using functions to generate the triangle shape for n lines where n is input by the user

9. The Fibonacci sequence is a series of numbers where every number is the sum of the two numbers before it. Here are the first few numbers: 0, 1, 1, 2, 3, 5, 8, 13... You must write the fib() function which takes in a positive integer, n, and returns the n-th

10. Fibonacci number. Your function must use any of the loops.

Output:

Example: n = 7, output = 8